Rati Devidze

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Experience

Present – July 2026 ***Reinforcement Learning & Machine Learning Algorithm Developer*** **- Applied Materials** (Germany)

- I apply Reinforcement Learning (RL) and Preference Learning to semiconductor fabrication (FAB) optimization to advance smart factories.

June 2026 – May 2024 ***Reinforcement Learning & Machine Learning Engineer*** **- minds.ai** – Remote (Germany)

- Implemented preference-learning-based policy optimization methods for semiconductor FAB scheduling, improving policy training efficiency and operational KPIs in industrial environments.
- Reduced environment state-space dimensionality to improve RL agent prediction quality, convergence behavior, and training stability.
- Developed and evaluated reinforcement learning pipelines for complex industrial optimization tasks.

May 2024 – Nov 2018 ***Ph.D. Researcher*** **- Max Planck Institute for Software Systems (MPI-SWS)** – Germany

- Conducted research on reward design, preference optimization, and reinforcement learning, resulting in publications at top-tier AI conferences: NeurIPS, ICML, IJCAI, and AAMAS.
- Built scalable experimentation pipelines for training, evaluation, and comparison of RL agents across simulated environments.
- Designed reward shaping and meta-learning methods for sparse-reward reinforcement learning settings.
- Led research projects in reinforcement learning, inverse reinforcement learning, and machine teaching.

May 2018 – Feb 2018 ***Research Immersion Lab*** **- Max Planck Institute for Software Systems (MPI-SWS)** – Germany

- Worked on iterative machine teaching in Markov Decision Processes under the supervision of Dr. Adish Singla and Dr. Parameswaran Kamalaruban.
- Implemented MDP solvers and conducted experimental evaluations of theoretical teaching algorithms.

Sep 2017 – July 2017 ***Research Immersion Lab*** **- CISPA Helmholtz Center for Information Security** – Germany

- Analyzed accountability properties of security protocols using the Tamarin theorem prover.

Aug 2016 – Mar 2014 ***Software Engineer*** **- Prodware Group & Akhali Kselebi Ltd** – Georgia

- Developed enterprise and telecommunications software systems using .NET, ASP.NET, and C#.
- Implemented backend services, analytics tools, and business applications.

Education

Apr 2024 – Nov 2018 ***Ph.D. in Computer Science*** **- Max Planck Institute for Software Systems (MPI-SWS)** – Germany

- Thesis: *Reward Design for Reinforcement Learning Agents*

Aug 2018 – Sep 2016 ***Graduate School of Computer Science*** **- Saarland University** – Germany

- Relevant coursework: Statistical Learning, Deep Learning, Information Retrieval, Data Mining

July 2016 – Sep 2012 ***B.Sc. in Computer Science*** **- Tbilisi State University** – Georgia

- GPA: 4.0/4.0; President's Scholarship Holder

Technical Skills

- **Programming:** Python, C#, SQL, Git, Linux
- **Machine Learning:** Reinforcement Learning, Preference Learning, RLHF, DPO, Offline RL, Deep Learning, Statistical Learning, Optimization
- **Reinforcement Learning Methods:** Reward Modeling, Reward Shaping, Policy Optimization, PPO, SAC, DQN, Meta-RL, Inverse RL, Sparse-Reward RL, Stable Baselines, RLLib
- **Frameworks and Tools:** PyTorch, NumPy, Pandas, Scikit-learn, Experiment Pipelines, Hyperparameter Optimization, Convex Optimization, CVXPY
- **Languages:** Georgian (native), English (professional), German (C1), Russian (A2)

Research Highlights

- **Reward Design for Reinforcement Learning:** Developed methods for designing informative and interpretable reward functions that accelerate RL convergence while improving explainability.
- **Sparse-Reward Reinforcement Learning:** Designed meta-gradient-based reward shaping approaches for improving exploration and learning efficiency in sparse-reward environments.
- **Machine Teaching and Inverse Reinforcement Learning:** Studied curriculum design and personalized demonstration strategies for sequential decision-making problems.

Selected Publications

My research has been published at leading machine learning conferences, including NeurIPS, ICML, IJCAI, and AAMAS. For a complete list of publications, kindly check my Google Scholar and DBLP.

- **NeurIPS 2022** – Exploration-Guided Reward Shaping for Reinforcement Learning under Sparse Rewards
- **NeurIPS 2021** – Explicable Reward Design for Reinforcement Learning Agents
- **AAMAS 2024** – Informativeness of Reward Functions in Reinforcement Learning
- **IJCAI 2019** – Interactive Teaching Algorithms for Inverse Reinforcement Learning

Teaching

- Teaching Assistant – Reinforcement Learning, MPI-SWS and Saarland University, 2021
- Teaching Assistant – Multi-Agent Reinforcement Learning Seminar, MPI-SWS and Saarland University, 2020
- Teaching Assistant – Machine Teaching Seminar, MPI-SWS and Saarland University, 2019

Awards & Scholarships

- Graduate School of Computer Science Scholarship, Saarland University.
- DAAD Scholarship Recipient for German Language Training at RWTH Aachen University.
- President's Scholarship Holder, Tbilisi State University.